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CS-499 Module 4 Milestone Four

Artifact Enhancement 3: Databases

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The artifact I selected for Milestone Four was the Travlr web application that I originally developed in CS-465. I selected this artifact because it already contained user authentication, trip management functionality, and a MongoDB database structure, making it an ideal candidate for a database-focused enhancement. While the application stored trip and user information, it did not include any relationship between users and the trips they wanted to book. This provided an opportunity to demonstrate database design, collection relationships, and authenticated data operations through a realistic reservation system.

To enhance the artifact, I implemented a MongoDB-backed reservation system that establishes relationships between authenticated users and trips. I created a new Reservation collection using Mongoose and designed a schema that references both the User and Trip collections through ObjectId relationships. The reservation schema includes reservation-specific information such as the number of guests, reservation status, and automatically generated timestamps. I also added indexing to improve query performance and support efficient retrieval of user reservations.

In addition to the database design changes, I developed a set of authenticated API endpoints that allow users to create, view, and delete reservations. The reservation controller uses Mongoose queries and populated references to retrieve trip information associated with each reservation. By using `populate()`, the application can display complete trip information without duplicating trip data inside reservation records. This approach demonstrates proper normalization and relationship management within a MongoDB database environment.

The public-facing website was also enhanced to integrate with the new database functionality. Users can now reserve trips directly from the Travel page, where each trip includes a Reserve button. When selected, the application creates a reservation linked to both the

authenticated user and the selected trip. The Reservations page was updated to retrieve reservation data from MongoDB and display the user's reservations, including trip details, reservation dates, guest counts, and status information. Users can also remove reservations through authenticated database operations.

This enhancement aligns with several CS-499 course outcomes. It demonstrates the ability to design and evaluate computing solutions using appropriate database technologies and software engineering practices. The project required designing a new database schema, implementing collection relationships, securing database operations through authentication, and integrating data across multiple application layers. The enhancement also demonstrates the ability to analyze requirements and develop a solution that improves both functionality and maintainability while preserving data integrity.

Through this enhancement, I gained a deeper understanding of database modeling and relationship management in MongoDB. I learned how to design collections that reference one another, how to use Mongoose population to retrieve related data, and how to implement secure user-specific database operations. I also gained practical experience integrating database functionality across the backend API layer and the frontend user interface.

One of the primary challenges I faced was designing a reservation system that fit naturally into the existing application architecture. Because the original application did not contain any relationships between users and trips, I needed to carefully design a new schema that maintained separation of concerns while supporting efficient queries. Another challenge involved integrating authenticated API operations with the public website while ensuring that users could only view and manage their own reservations. Addressing these challenges strengthened my understanding of database design, data relationships, and secure application development within a full-stack environment.